

THE LIFE CYCLE DESIGN

- Do you know what this symbol means?



- Can you give examples of environmental impact?

A cost-benefit analysis involves adding up the benefits of an action and then comparing these with the costs associated with it. The results of the analysis are often expressed as a payback period – this is the time it takes for benefits to repay costs.

■ Materials selection criteria

Careful selection of **environmentally sustainable** building materials is the easiest way to begin incorporating sustainable design principles in buildings. Traditionally, price has been the main consideration when comparing similar materials, or materials designated for the same function, with no consideration of social or environmental costs.

A detailed analysis of building products, from collecting **raw materials** to their ultimate **disposal**, provides a better understanding of the long-term costs of materials.

Following the principles of **life cycle design**, each step of the **manufacturing process**, from gathering raw materials, manufacturing, distribution, and installation, to ultimate reuse or disposal, is examined, looking for its **environmental impact**.

■ Phases in the life of a material

A material's life cycle can be organised into three phases: **pre-building**, **building** and **post-building**. The evaluation of building materials' environmental impact at each stage allows for a cost-benefit analysis over the lifetime of a building.

- The **pre-building phase** describes the production and **delivery** process of a material up to – but not including –

the point of installation. This includes discovering raw materials in nature as well as extracting, manufacturing, packaging and transporting to the building site. This phase has the most potential for causing environmental damage. The ecological damage related to the use of natural resources and their conversion into building materials includes loss of **wildlife habitat**, erosion, and water and air pollution.

- The **building phase** refers to the useful life of a building material. This phase begins with the material **assembly** into a structure, includes the maintenance and repair of the material, and extends throughout the life of the material as part of the building. The selection of building materials for reduced construction waste, and waste that can be recycled, is critical in this phase of the building life cycle. **Long-term exposure** to certain building materials may be hazardous to the health of a building's occupants.
- The **post-building phase** refers to the building materials when their usefulness in a building has expired. At this point, a material may be completely reused, have its components recycled back into other products, or be thrown away.

assembly: *assemblaggio*
delivery: *consegna*
disposal: *smaltimento*
environmentally sustainable: *sostenibile per l'ambiente*
long-term exposure: *esposizione continua*
manufacturing: *produzione*
raw material: *materiale grezzo*
wildlife habitat: *habitat naturale*



Green building
in Malaga, Spain

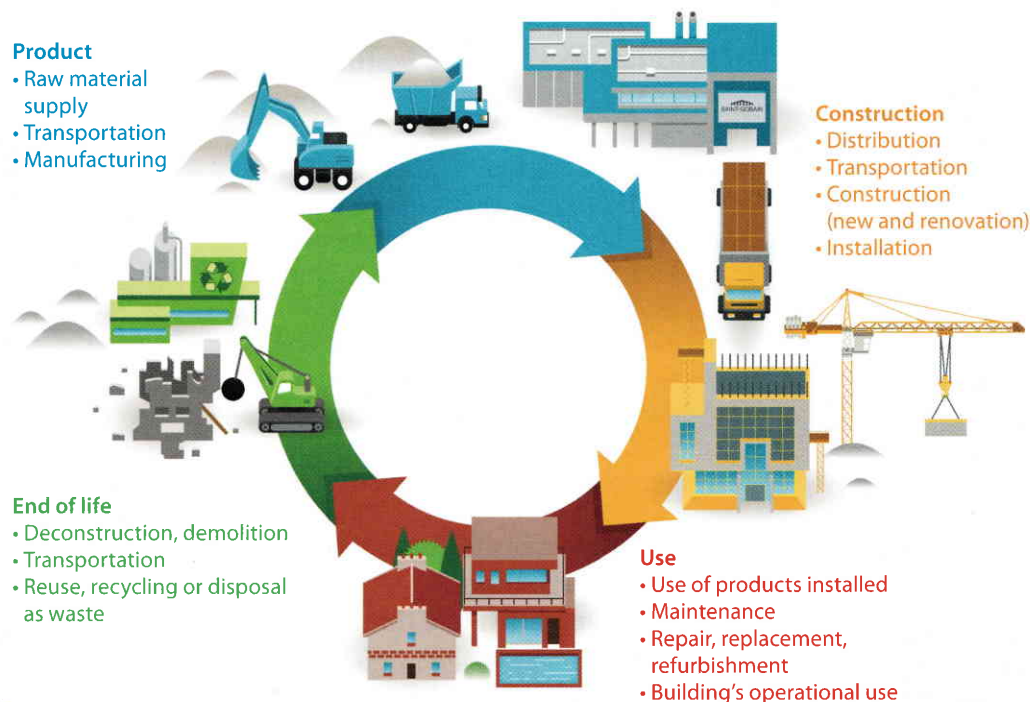
- 4  Look at the diagram on the right. It represents the Life Cycle Assessment of a construction product. Describe it orally, with the help of what you have just read in the reading passage.



LCA INTERNATIONAL STANDARDS

Results obtained via an LCA can be trusted as they are based on international standards, ISOs 14040 and 14044. This helps to ensure consistency, reliability and independence from commercial pressure.

THE LCA OF A CONSTRUCTION PRODUCT



- 5  ^{2.7} Dr Pastakkaya, a Turkish researcher who worked with fellow academics in Nottingham, UK, on research entitled *New Approaches to Eco House Design and Renewable Energy Applications in Sustainable Buildings*, is talking about *eco-house design*. Listen and decide if the statements below are true or false. Correct the false ones.

- | | | |
|---|-----|---|
| 1. Eco design principles used by the ancient civilizations cannot be used today. | T F | ☐ ☐ |
| 2. The Greeks placed their windows in order to catch more wind. | T F | ☐ ☐ |
| 3. Houses in Egypt and Persia had devices to exploit natural ventilation. | T F | ☐ ☐ |
| 4. Long before the Romans, houses had walls that could help to make living spaces cooler. | T F | ☐ ☐ |
| 5. Hypocausts are systems used to make a room warmer. | T F | ☐ ☐ |
| 6. Thermal comfort is one of the main requirements in a house. | T F | ☐ ☐ |
| 7. At the beginning of the 20 th century, systems to provide hot water were simple and efficient. | T F | ☐ ☐ |
| 8. Solar-powered systems came onto the market in the 1930s. | T F | ☐ ☐ |
| 9. Oil production made renewable energy applications more popular. | T F | ☐ ☐ |
| 10. The cost of fossil fuels and environmental problems are the reasons for the comeback of eco-technologies. | T F | ☐ ☐ |

- 6  Find words in the text to match the following synonyms or definitions.

1. poisonous	
2. lack of humidity	
3. existing for a long time	
4. ensure (v.)	
5. gratifying the senses	

CHOOSING BUILDING MATERIALS

- Underline the materials which you think are not used in bio-architecture: bamboo, brick, cellulose fibre, clay, cork, juta, steel, stone, timber.

Make a list of materials inside your house which you think may be recyclable.

Passive solar orientation – south-facing windows for light and heat, and minimum north-facing to reduce heat loss – is one of the characteristics of an eco-house.

brick: *mattone*
cabinetry: *mobili*
clay: *argilla*
cork: *sughero*
fixture: *infixso*
glue: *colla*
timber: *legname*
wax: *cera*

■ Building materials features

Building materials have the purpose of **making the place where we live safe and comfortable**. They satisfy our desire for beauty, respond to technical needs but also influence our health and the way we feel inside our homes.

These materials should be eco-friendly, sustainable and have to be chosen paying great attention to any negative effect on the environment, such as production of toxic gases, water pollution, soil pollution.

■ Where are building materials employed?

They can be sorted into three groups:

- materials employed in the **structure** of the building (foundations, walls, attic, etc.);
- materials used to improve the **performances** of the building (insulations, protections, **fixtures**);
- finishing materials** which complete the structure of the building and are the ones which come into contact with the people living in it (paints, **glues**, floors, etc.).

In the process of building a house **each material used should interact with the others** and contribute to the building's global performance. While it may be quite simple to identify and select eco-friendly materials, it is often more complicated to understand how one may



interact with others. Natural materials can be employed within an inadequate context and this can make their use pointless or even harmful.

■ The power of eco-building materials

To sum up, **eco-materials should:**

- be biodegradable and recyclable
- minimize the effects of toxic and hazardous substances on health
- contribute to create a more sustainable environmental future.

They can do this by:

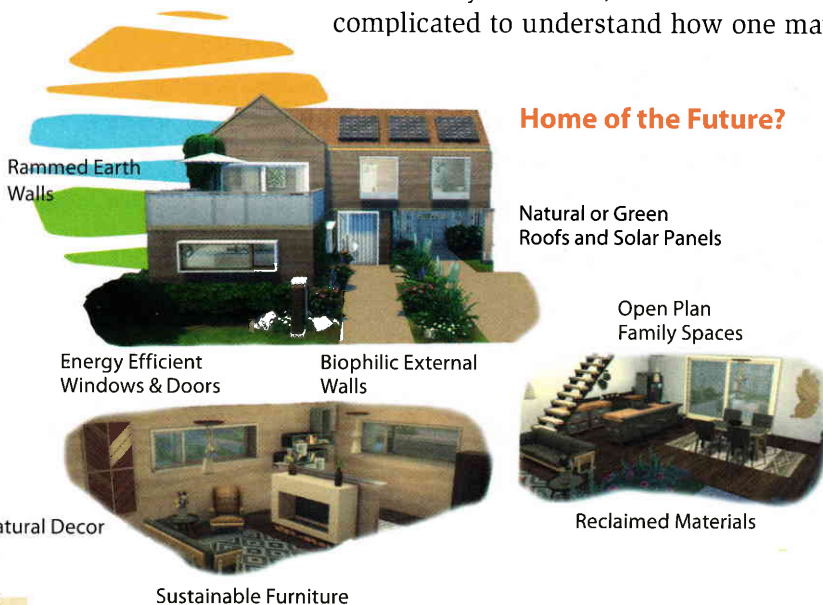
- minimising their impact on the environment
- avoiding or reducing dependence on non-renewable energy sources
- increasing indoor air quality
- increasing the efficiency of resources
- avoiding or reducing problems connected to allergies.

■ Benefits in the long term

Encouraging the use of such materials will therefore have several **social and environmental benefits** as well as providing quality buildings and manufacturing products.

Among the materials which can be used in bio-architecture, we can find **timber**, **brick**, **stone**, **clay** (for the structures), **cork**, **juta**, **cellulose fibre** (used as insulating materials), **bamboo** (for flooring and **cabinetry**), **natural paints** and **natural waxes** (for finishing).

Home of the Future?



7 Decide if the following statements are true or false and correct the false ones.

T F

1. Building materials have the main purpose of satisfying our desire for beauty. ☐ ☐
2. Bio-architecture building materials have to respect the environment as much as possible, from their production to their use. ☐ ☐
3. As well as choosing natural materials, it is important to consider how each material employed interacts with others. ☐ ☐
4. There are either good or bad materials. ☐ ☐
5. In bio-architecture, the possibility of recycling a material is an important fact to determine its choice. ☐ ☐
6. Cork is an example of eco-friendly material used in building structures. ☐ ☐

9

8 Complete these notes about eco-materials.

• Used for	a. b. c. finishing
• They should:	
• They can contribute to the protection of the environment because they:	 increase the efficiency of resources



GRAMMAR

PRESENT PERFECT SIMPLE vs PRESENT PERFECT PROGRESSIVE

Present Perfect Simple

FORM

- AFFIRMATIVE SENTENCES
subject + *have/has* + past participle
- NEGATIVE SENTENCES
subject + *have/has* + *not* + past participle
- INTERROGATIVE SENTENCES
have/has + subject + past participle

USE

1. Present impact of past action/recent event
➤ *The town is changing because the council **has started** improving the quality of new buildings.*
2. Unfinished action started in the past
➤ *The restoration project **hasn't progressed** this month. (from the first day of the month till now)*
3. Unfinished periods of time
➤ *Mrs Hollister **has worked** as a builder for 20 years. (he's still working)*
4. Present result of completed action in a period of time
➤ *The workers **have painted** three rooms since this morning.*
5. Finished past events with no time reference
➤ *The building **has undergone** many changes.*

TIME EXPRESSIONS

just, ever, never, already, yet, for (+ total duration), since (+ beginning of duration), still, so far, recently, this/these + time expression, today, how long
➤ *How long **have you worked** on the project of the eco-house?*
– *For one year.*

Present Perfect Progressive

FORM

- AFFIRMATIVE SENTENCES
subject + *have/has* + *been* + base form + -ing
- NEGATIVE SENTENCES
subject + *have/has* + *not* + *been* + base form + -ing
- INTERROGATIVE SENTENCES
have/has + subject + *been* + base form + -ing

USE

1. Unfinished continuous or repetitive actions
➤ *The baby **has been crying** for twenty minutes.*
2. Recently finished actions with evidence
➤ *I am tired, I **have been painting** the walls with this natural paint for hours.*
3. Unfinished activities in a period of time
➤ *They **have been assembling** the structure all day. (both day and assembling haven't finished yet)*

TIME EXPRESSIONS

for (+ total duration), since (+ beginning of duration), recently, this/these + time expression, today, how long
➤ *Dad **has been painting** the baby's room **since** two o'clock.*



9 Complete the following sentences with the Present Perfect Simple or the Present Perfect Progressive of the verb given.

1. It's 10.30 already, but the workers (not started) work yet this morning.
2. It (rain) all week and the excavation is still half-done.
3. We (be) to the building site three times this week.
4. Look at dad's hair! He (paint) the ceiling.
5. I am exhausted, I (wax) the wooden floors since 8 this morning.
6. That architect (project) a state-of-the-art eco-building recently.
7. Susan's allergy is improving now because she (start) using eco-products for her house.
8. you (have) troubles in converting your house to solar power?
9. I (met) my friend at a refresher course in eco-architecture today.
10. Oh dear, your house (get) greener!